87 Prince St, Mullumbimby: deck and awning footings

For **Old Mill, Mullumbimby** · Bryce Noakes · 2 Nov 2026

Design reviewed by **Ollie Harris, structural engineer** (markup 2 Nov 2026). Everything below follows his markup; anything he didn't flag is as drawn and approved.

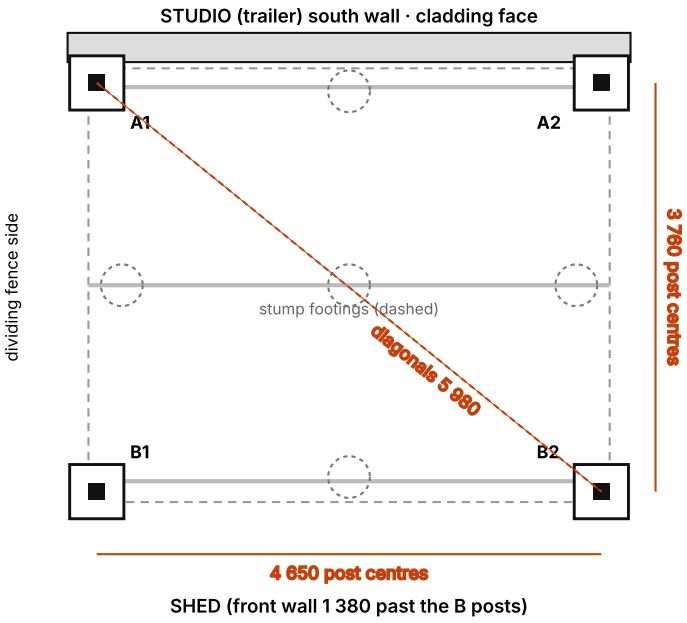
3D model, plans and engineering

TODAY: POUR THE 4 POST FOOTINGS WITH CAST-IN BRACKETS

Item	Spec (engineer)	Qty	Old Mill: have it / sub
Cast-in post bracket (stirrup)	Hot-dip galvanised, cast-in, to suit a **150 × 150 post**, M12 bolt. Must fit inside a 500 deep footing with concrete under the end and the post foot just above ground.	4	
Concrete, post footings	20 kg bags. Four footings **500 × 500 × 500** = 0.5 m³ (about 54 bags), plus spare	60	
M12 bolts for the brackets	HDG, about M12 × 200 with nuts and washers, to suit 150 post + bracket plates (needed when posts go in)	4	
Posts (to check the bracket fit)	**150 × 150 H4** treated pine, 3.6 m	4	

Optional today, if pouring the stump footings too: 5 × Pryda Post Anchor High Wind 600 × 90 M12, and about 40 more bags for five 400 dia × 525 deep holes.

FOOTING SETOUT



POST CENTRES (MM)

Post	Along the studio wall, from its west end	Out from the studio cladding
A1	99	146
A2	4 749	146
B1	99	3 906
B2	4 749	3 906

Check: A1–A2 and B1–B2 = 4 650; A–B = 3 760; both diagonals = 5 980. Each hole is 500 square, centred on its post. The A holes come within about 100 of the studio cladding: keep clear of the trailer stands.

BRACKET SETTING (ENGINEER'S DETAIL)

- Centre the bracket in the hole; post foot just above ground.
- **Orientation:** bracket plates **parallel to the studio wall** (the bearer line), so the bolt runs towards the shed, the same way as the knee-brace bolts.
- Plates plumb and square to the setout lines; prop them until the concrete sets.
- Footings 500 × 500 × 500.

Before you pour: check the bracket's stem fits a 500 deep hole with concrete under the end. A 600 long high-wind anchor won't; if that's all that's available, ring Ollie before digging deeper (650 deep would take about 75 bags instead of 60).

Full material list for quote: deck and awning

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TIMBER

Item	Spec	Length	Qty	Old Mill price / sub
Posts	150 × 150 H4 treated pine, on cast-in brackets	3.6 m	4	
Roof beams B1 and B2	**240 × 45 H3 MGP12** (2 plies per beam, one each side of the posts)	6.0 m	4	
Rafters	190 × 45 H3 MGP10, 6 at 930 centres	6.0 m	6	
Fascia	140 × 35 H3	6.0 m	1	
Purlins	70 × 35 H3	6.0 m	8	
Deck bearers	190 × 45 H3 MGP10, continuous: **single** on lines A and B, **double** on line C	4.8 m	4	
Deck joists	140 × 45 H3, continuous over the span, max 450 centres (engineer: keep the bigger joists for the laundry and kitchen)	4.8 m	13	
Knee braces	Min 120 × 45 H3, at all 4 corner posts, M12 bolt each end	4.8 m	1	
Stumps	90 × 90 F7 H4 (makes all five)	2.7 m	1	
Decking	90 × 22 H3 pine	5.4 m	42	
Screen battens	70 × 45 H3, on edge (vertical screen, rails, hangers)	4.8 m	25	

HARDWARE AND CONCRETE

Item	Spec	Qty	Old Mill price / sub
Cast-in post brackets	HDG, for 150 × 150 posts, M12 (see page 1)	4	
Stump anchors	Pryda Post Anchor High Wind 600 × 90 M12	5	
Bolts	M12 HDG: beams to posts, bearers to posts and stumps, 8 for the knee braces, bracket bolts; 50 × 50 square washers	1 lot	
Straps and screws	Triple grips or cyclone straps (every rafter to the beams), batten and decking screws	1 lot	
Concrete	20 kg bags: 60 for the post footings, 40 for the stump footings	100	
End sealer	For cut ends of treated timber	1	

Roof sheets (Zincalume and clear corrugated) are not in this list.

ENGINEER'S CHANGES (OLLIE HARRIS, 2 NOV 2026)

- Bearer lines A and B: **single 190 × 45 MGP10, continuous**, inside face of the posts.
- Bearer line C: **double 190 × 45 MGP10, continuous**, on 3 stumps.
- All bearers continuous and supported by an intermediate post.
- Joists continuous over the span, max 450 centres; keep the bigger size.
- **Diagonal timber knee brace** at all 4 corner posts: min 120 × 45, M12 bolt each end.
- Posts on **cast-in brackets**, oriented as drawn; footings **500 × 500 × 500**.
- B1 and B2: **2/240 × 45 MGP12**.
- Rafters 190 × 45 at 930 and fascia 140 × 35: approved as drawn.

For Old Mill: if you can't get 240 × 45 MGP12 H3, 150 × 150 H4 posts or 150 cast-in brackets, note the substitute in the right-hand column and I'll check it with the engineer before ordering. Plans, 3D model and the engineer's markup: **tradearts.work/prince-street-deck-awning/p25.html**